

SKILL-1

```
ritwik@DESKTOP-PK3LAMN:~$ mkdir -p ~/ossp_project
cd ~/ossp_project
ritwik@DESKTOP-PK3LAMN:~/ossp_project$ pwd
/home/ritwik/ossp_project
ritwik@DESKTOP-PK3LAMN:~/ossp_project$ mkdir src
mkdir bin
mkdir build
ritwik@DESKTOP-PK3LAMN:~/ossp_project$ tree
Command 'tree' not found, but can be installed with:
sudo snap install tree # version 2.1.3+pkg-5852, or
sudo apt install tree # version 2.3.1-1
See 'snap info tree' for additional versions.
ritwik@DESKTOP-PK3LAMN:~/ossp_project$ sudo apt install tree -y
[sudo: authenticate] Password:
Installing:
  tree

Summary:
  Upgrading: 0, Installing: 1, Removing: 0, Not Upgrading: 162
  Download size: 53.5 kB
  Space needed: 125 kB / 1024 GB available

Get:1 http://archive.ubuntu.com/ubuntu resolute/universe amd64 tree amd64 2.3.1-1 [53.5 kB]
Fetched 53.5 kB in 1s (62.4 kB/s)
Selecting previously unselected package tree.
(Reading database ... 42490 files and directories currently installed.)
Preparing to unpack ../tree_2.3.1-1_amd64.deb ...
Unpacking tree (2.3.1-1) ...
Setting up tree (2.3.1-1) ...
Processing triggers for man-db (2.13.1-1build1) ...
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```
ritwik@DESKTOP-PK3LAMN:~/ossp_project$ nano src/process.c
ritwik@DESKTOP-PK3LAMN:~/ossp_project$ gcc src/process.c -o bin/process
ritwik@DESKTOP-PK3LAMN:~/ossp_project$ ls -l bin
total 16
-rwxr-xr-x 1 ritwik ritwik 16296 Sep 16 04:00 process
ritwik@DESKTOP-PK3LAMN:~/ossp_project$ ./bin/process
Parent Process Started
Parent PID: 790

Parent waiting for child...

Child Process Started
Child PID: 791
Parent PID: 790
Child executing ls command...
total 12
drwxr-xr-x 2 ritwik ritwik 4096 Sep 16 04:00 bin
drwxr-xr-x 2 ritwik ritwik 4096 Sep 16 03:59 build
drwxr-xr-x 2 ritwik ritwik 4096 Sep 16 04:00 src
Child process completed.
Parent process exiting.
ritwik@DESKTOP-PK3LAMN:~/ossp_project$ nano Makefile
ritwik@DESKTOP-PK3LAMN:~/ossp_project$ make
gcc -Wall -Wextra src/process.c -o bin/process
```

```

ritwik@DESKTOP-PK3LAMN:~/ossp_project$ make run
./bin/process
Parent Process Started
Parent PID: 827

Parent waiting for child...

Child Process Started
Child PID: 828
Parent PID: 827
Child executing ls command...
total 16
-rw-r--r-- 1 ritwik ritwik 167 Sep 16 04:01 Makefile
drwxr-xr-x 2 ritwik ritwik 4096 Sep 16 04:01 bin
drwxr-xr-x 2 ritwik ritwik 4096 Sep 16 03:59 build
drwxr-xr-x 2 ritwik ritwik 4096 Sep 16 04:00 src
Child process completed.
Parent process exiting.
ritwik@DESKTOP-PK3LAMN:~/ossp_project$ make clean
rm -f bin/process
ritwik@DESKTOP-PK3LAMN:~/ossp_project$ git init

```

```

ritwik@DESKTOP-PK3LAMN:~/ossp_project$ git init
hint: Using 'master' as the name for the initial branch. This default branch name
hint: will change to "main" in Git 3.0. To configure the initial branch name
hint: to use in all of your new repositories, which will suppress this warning,
hint: call:
hint:
hint:   git config --global init.defaultBranch <name>
hint:
hint: Names commonly chosen instead of 'master' are 'main', 'trunk' and
hint: 'development'. The just-created branch can be renamed via this command:
hint:
hint:   git branch -m <name>
hint:
hint: Disable this message with "git config set advice.defaultBranchName false"
Initialized empty Git repository in /home/ritwik/ossp_project/.git/
ritwik@DESKTOP-PK3LAMN:~/ossp_project$ tree
.
├── Makefile
├── bin
├── build
├── src
│   └── process.c
└──

4 directories, 2 files
ritwik@DESKTOP-PK3LAMN:~/ossp_project$ git status
On branch master

No commits yet

Untracked files:
  (use "git add <file>..." to include in what will be committed)
    Makefile
    src/

nothing added to commit but untracked files present (use "git add" to track)

```

```

ritwik@DESKTOP-PK3LAMN:~/ossp_project$ ls src
process.c
ritwik@DESKTOP-PK3LAMN:~/ossp_project$ gcc src/process.c -o bin/process
ritwik@DESKTOP-PK3LAMN:~/ossp_project$ ls -l bin
total 16
-rwxr-xr-x 1 ritwik ritwik 16296 Sep 16 04:03 process
ritwik@DESKTOP-PK3LAMN:~/ossp_project$ ./bin/process
Parent Process Started
Parent PID: 953

Parent waiting for child...

Child Process Started
Child PID: 954
Parent PID: 953
Child executing ls command...
total 16
-rw-r--r-- 1 ritwik ritwik 167 Sep 16 04:01 Makefile
drwxr-xr-x 2 ritwik ritwik 4096 Sep 16 04:03 bin
drwxr-xr-x 2 ritwik ritwik 4096 Sep 16 03:59 build
drwxr-xr-x 2 ritwik ritwik 4096 Sep 16 04:00 src
Child process completed.
Parent process exiting.
ritwik@DESKTOP-PK3LAMN:~/ossp_project$ git status
On branch master

No commits yet

Untracked files:
  (use "git add <file>..." to include in what will be committed)
    Makefile
    bin/
    src/

nothing added to commit but untracked files present (use "git add" to track)

```

```

ritwik@DESKTOP-PK3LAMN:~/ossp_project$ git add .
ritwik@DESKTOP-PK3LAMN:~/ossp_project$ git status
On branch master

No commits yet

Changes to be committed:
  (use "git rm --cached <file>..." to unstage)
    new file:   Makefile
    new file:   bin/process
    new file:   src/process.c

ritwik@DESKTOP-PK3LAMN:~/ossp_project$ git commit -m "Implement process abstraction using fork and exec"
[master (root-commit) 8f02d02] Implement process abstraction using fork and exec
3 files changed, 61 insertions(+)
create mode 100644 Makefile
create mode 100755 bin/process
create mode 100644 src/process.c
ritwik@DESKTOP-PK3LAMN:~/ossp_project$ git log --oneline
8f02d02 (HEAD -> master) Implement process abstraction using fork and exec
ritwik@DESKTOP-PK3LAMN:~/ossp_project$ |

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